



Key terms & glossary

The exact wording examiners expect — the five ethical concepts, the six sustainability principles, value systems, Earth's systems and circular-economy thinking.

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1 The five ethical concepts

Learn the exact wording — examiners reward the precise term, not a paraphrase.

Beneficence: Maximising benefits and minimising risks/harms to people, animals and the environment.

Integrity: Honest, accurate reporting and dissemination of results, including unfavourable ones.

Justice: Fair distribution of benefits, risks, costs and resources; no group unfairly burdened.

Non-maleficence: Avoiding causing harm; where harm is unavoidable, it must not be disproportionate to the benefit.

Respect: Consideration of the welfare, beliefs, rights and autonomy of those affected, including future generations and ecosystems.

2 The three ethical approaches

Three lenses for applying the concepts to a decision.

Consequentialism

An action is judged right by its **outcomes** — the greatest good / least harm overall.

Deontology

Judged by **duties and rules** — some acts are right or wrong regardless of consequences.

Virtue ethics

Judged by the **character** of the moral agent — what a person of good character would do.

3 Value systems

How a person's worldview shapes every environmental decision.

Anthropocentrism

Human-centred. Nature has **instrumental** value — worth for the benefits it gives people.

Biocentrism

Life-centred. Every living organism has **intrinsic** value (atomistic — counts individuals).

Ecocentrism

System-centred. The whole ecosystem, living and non-living, has intrinsic value (holistic).

Technocentrism

Technology-centred. Faith that science and engineering can solve environmental problems.

BIO VS ECO

Biocentrism values individual living things; ecocentrism values the whole system, including non-living parts. Keep “individuals” vs “whole web” crisp.

4 The six sustainability principles

The decision criteria that run across all four outcomes.

Principle	Plain meaning
Conservation of biodiversity & ecological integrity	Maintain the variety of life and keep ecosystems functioning so natural processes and services continue.
Efficiency of resource use	Get the most benefit from the least resource, with the least waste.
Intergenerational equity	Fairness between generations — don't compromise the ability of future generations to meet their needs.
Intragenerational equity	Fairness within the present generation — across people and places alive now.
Precautionary principle	Where harm could be serious or irreversible, lack of full certainty is not a reason to delay action.
User-pays principle	Those who use a resource or cause an impact bear its environmental cost (the polluter pays).

THE PREFIX IS THE ANSWER

Inter generational = across time (future generations); intra generational = within today's generation.

5 Earth's four interrelated systems

Every environmental impact lands in one or more of these — and cascades between them.

Atmosphere

The envelope of gases — air, greenhouse gases, water vapour, heat and weather.

Biosphere

All living things and the zones they occupy.

Hydrosphere

All water — oceans, rivers, lakes, groundwater, ice.

Lithosphere

The solid Earth — rock, soil, landforms and minerals.

6 Circular economy & related terms

Linear economy

Take → make → use → dispose. Two open ends: finite resources in, waste and pollution out.

Circular economy

Designed so resources are kept in use as long as possible and “waste” becomes an input — make → use → return. Recycling is the last resort, not the whole idea.

◇ Quick glossary

Term	Definition
Sustainable development	Development that meets present needs without compromising future generations' ability to meet theirs (Brundtland, 1987).
Ecosystem services	Benefits ecosystems provide: provisioning, regulating, supporting and cultural.
Endemic	A species found naturally in one region and nowhere else.
Intrinsic vs instrumental value	Worth in/of itself vs worth as a means to a human end.
Mitigation vs adaptation	Reducing net emissions (the cause) vs building resilience to unavoidable change (the effects).
Stakeholder	Any party affected by, or able to influence, an environmental decision.

★ Self-check

Distinguish between intergenerational and intragenerational equity. (2 marks)

Weak: They're both about being fair to people with the environment.

Strong: Intergenerational equity is fairness between generations — not depleting resources so future generations can still meet their needs. Intragenerational equity is fairness within the present generation — distributing environmental benefits and costs fairly across people and places alive today.

Name the five ethical concepts examined in VCE Environmental Science. (2 marks)

Weak: Being good, honest and fair, and not hurting things.

Strong: Beneficence, integrity, justice, non-maleficence and respect.



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